

Setting CHARGE MODE for the x743 modules

1. Variable definition

```
CAEN_DGTZ_DPP_X743_Event_t  *ChargeEvents[MAX_NB_OF_CHANNELS_IN_SYSTEM];
CAEN_DGTZ_DPP_X743_Params_t  ChargeParams;
int                          NbOfEventsPerChannel[MAX_NB_OF_CHANNELS_IN_SYSTEM];

uint32_t                    ReadoutBufferSize;
uint32_t                    *ReadoutBuffer;
uint32_t                    ReadoutBufferMaxSize;
uint32                      NumberOfEvents ;
```

2. Enable /Disable Charge Mode

2.1 Enable Charge Mode + Allocation DPP Events

```
CAEN_DGTZ_SetSAMAcquisitionMode(DeviceHandle,CAEN_DGTZ_AcquisitionMode_DPP_CI);
```

Allocating the DPP Events :

```
CAEN_DGTZ_MallocDPPEvents(DeviceHandle, (void **)ChargeEvents, &allocatedSize);
```

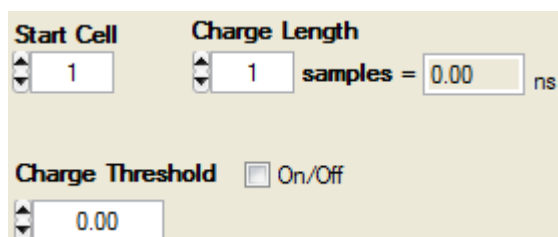
2.2 Disable Charge Mode + Free of DPP Events

```
CAEN_DGTZ_SetSAMAcquisitionMode(DeviceHandle,CAEN_DGTZ_AcquisitionMode_STANDARD);
```

Free DPP Events :

```
CAEN_DGTZ_FreeDPPEvents(DeviceHandle, (void **)ChargeEvents);
```

3. Setting Charge Parameters



After building the ChargeParams Struct (cf. x743 documentation), use :

```
CAEN_DGTZ_SetDPPParameters(DeviceHandle,dummy, &ChargeParams);
```

4. Allocation of Readout Buffer : **once at the start of the acquisition or Software Startup**

Same as for the Standard Mode:

```
CAEN_DGTZ_MallocReadoutBuffer(DeviceHandle, (char **)& ReadoutBuffer,  
&ReadoutBufferMaxSize);
```

5. Reading Event Buffer

Same as for the Standard Mode:

```
CAEN_DGTZ_ReadData(DeviceHandle, CAEN_DGTZ_SLAVE_TERMINATED_READOUT_MBLT, (char  
*)ReadoutBuffer, &ReadoutBufferSize);
```

6. Decoding Charge Event

If ChargeMode is enabled, instead of using CAEN_DGTZ_DecodeEvent for the standard mode, use the function :

```
CAEN_DGTZ_GetDPPEvents(DeviceHandle, (char *)ReadOutBuffer, ReadoutBufferSize, (void  
**)ChargeEvents, NbOfEventsPerChannel);
```

7. Free of the Readout Buffer : **once at the end of the acquisition or at Software Exit**

Same as for Standard Mode:

```
CAEN_DGTZ_FreeReadoutBuffer((char **)&ReadoutBuffer);
```